

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273506

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

LI; Jian et al.

SUBSTRATE SUPPORT WITH SENSOR

Abstract

A substrate support assembly is provided including: a shaft; and a substrate support including: a substrate support body attached to the shaft, the substrate support body formed of a first material; and a tube positioned inside the substrate support body, the tube formed of a second material having a melting point that is higher than a sintering temperature of the first material.

Inventors: LI; Jian (Fremont, CA), ROCHA-ALVAREZ; Juan Carlos (San Carlos, CA), SUN; Jennifer Y. (Mountain View, CA)

Applicant: Applied Materials, Inc. (Santa Clara, CA)

Family ID: 1000007715775

Appl. No.: 18/586104

Filed: February 23, 2024

Publication Classification

Int. Cl.: H01L21/687 (20060101); H01L21/67 (20060101)

U.S. Cl.:

CPC H01L21/68785 (20130101); H01L21/67248 (20130101); H01L21/68757 (20130101);

Background/Summary

BACKGROUND

Field

[0001] Embodiments of the present disclosure generally relate to improved components (e.g., substrate supports) for use in process chambers, such as semiconductor process chambers, and

methods of manufacturing the improved components.

Description of the Related Art

[0002] Substrate supports in process chambers (e.g., semiconductor process chambers) often include one or more sensors to perform measurements used for controlling the process (e.g., deposition) performed in the process chamber. For example, substrate supports often include one or more temperature sensors inside the substrate support. In some embodiments, the temperature sensors can be used to control heat provided to the substrate support. The temperature sensors (e.g., thermocouples) are connected to one or more wires or leads routed through the interior of the substrate support and through a shaft that can be used to rotate the substrate support. Positioning a sensor in the interior of the substrate support can be challenging and/or expensive, especially when the sensor is to be located near the outer edge of the substrate support inside the substrate support. Current methods generally include forming separate plates with cavities to be used for placement of the sensor and then diffusion bonding the plates together. Diffusion bonding can be an expensive process.

[0003] Accordingly, improved substrate supports and processes for forming improved substrate supports with one or more sensors positioned inside the substrate support and near the outer edge of the substrate support is needed.

SUMMARY

[0004] Embodiments of the present disclosure generally relate to improved components (e.g., substrate supports) for use in process chambers, such as semiconductor process chambers, and methods of manufacturing the improved components.

[0005] In one embodiment, a substrate support assembly is provided comprising: a shaft; and a substrate support comprising: a substrate support body attached to the shaft, the substrate support body formed of a first material; and a tube positioned inside the substrate support body, the tube formed of a second material having a melting point that is higher than a sintering temperature of the first material.

[0006] In another embodiment, a method for forming a substrate support is provided comprising: embedding a tube in a sintering material, an interior of the tube substantially devoid of the sintering material; performing a sintering process on the sintering material with the tube embedded in the sintering material to form a substrate support body; and inserting a sensor into the interior of the tube after performing the sintering process.

[0007] In another embodiment, a method for forming a substrate support is provided comprising: forming a first plate by performing a first sintering process on a first sintering material, the first plate having a first side and an opposing second side, wherein the first sintering process partially sinters the first sintering material; forming a second plate by performing a second sintering process on a second sintering material, the second plate having a first side and an opposing second side, wherein the second plate includes a recess on the first side and the second sintering process partially sinters the second sintering material; placing a powder inside the recess; positioning the first plate on top of the first side of the second plate, wherein a region between the first plate and recess of the second plate forms a cavity; and performing a third sintering process on the first plate and the second plate to join the first plate to the second plate and form a substrate support body having a top, a bottom, and one or more sides connecting the top to the bottom, wherein the powder is formed of a material having a higher melting point than a temperature of the sintering material during the third sintering process to join the first plate to the second plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] So that the manner in which the above recited features of the present disclosure can be

understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

[0009] FIG. 1 shows a processing system, according to one embodiment.

[0010] FIGS. 2A-2C show different stages of a method for forming the substrate support assembly from FIG. 1, according to one embodiment.

[0011] FIG. 3 is a process flow diagram of a method for forming the substrate support assembly of FIG. 1, according to one embodiment.

[0012] FIGS. 4A-4D show different stages of forming an alternative substrate support assembly, according to one embodiment.

[0013] FIG. 5 is a process flow diagram of a method for forming the alternative substrate support assembly, according to one embodiment.

[0014] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0015] Embodiments of the present disclosure generally relate to improved substrate supports or other components for use in process chambers, such as semiconductor process chambers, and improved methods of manufacturing the improved components. The improvements generally relate to forming a void before sintering the material (e.g., aluminum nitride) that forms the body of the substrate support. In one embodiment, a tube is embedded in sintering material before a sintering process is performed. The tube can be formed of a material (e.g., molybdenum) having a melting point that is higher than the temperature at which the sintering process is performed. The interior of the tube can then be easily accessed after the sintering process by drilling a hole from the bottom of the substrate support to the interior of the tube. A flexible sensor can then easily be positioned in the tube to underlie an edge region of the substrate during processing. In another embodiment, two partially sintered plates are joined together in an additional sintering process to form a substrate support. A recess on a side of one of the partially sintered plates is used to form a cavity between the two plates when the two plates are sintered together. The cavity can then be easily accessed after the sintering process by drilling a hole from the bottom of the substrate support to the interior of the cavity. A flexible sensor can then easily be positioned in the cavity to underlie an edge region of the substrate during processing.

[0016] By having the void in the tube or the cavity accessible for positioning a sensor after the substrate support body is sintered in these two embodiments, the additional step of diffusion bonding sintered components (e.g., plates) is avoided. Conventional practice has been to fully sinter two separate plates and then join the plates together in a diffusion bonding process, where the plates are configured to form a void for insertion of a sensor after the diffusion bonding process. Diffusion bonding can be an expensive and time consuming process. The methods disclosed herein avoid this costly diffusion bonding process for forming the substrate support body.

[0017] Although the following disclosure mainly describes improved methods for forming substrate supports with internal sensors, the methods can be more generally applied to form other components used in process chambers, such as electrostatic chucks, showerheads, or other components that are exposed to harsh process conditions while including internal devices, such as sensors.

[0018] FIG. 1 shows a processing system **100**, according to one embodiment. The processing system **100** includes a process chamber **101**, a gas supply system **114**, and a vacuum pump **126**. The process chamber **101** includes a chamber body **102** enclosing an interior volume **104**. In one

embodiment, the process chamber **101** is a deposition chamber. In other embodiments, the process chamber **101** can include additional components (not shown) to perform other processes, such as a plasma deposition or etching process.

[0019] The process chamber **101** can include a showerhead **110** for directing process gases into the interior volume **104** of the process chamber **101**. The vacuum pump **126** can be used to exhaust gases from the interior volume **104** and to maintain a specified pressure in the interior volume **104** during processing.

[0020] The process chamber **101** further includes a substrate support assembly **200** having a substrate support body **210** positioned in the interior volume **104**. A substrate **50** can be positioned on the substrate support body **210** during processing, such as a deposition. The substrate support assembly **200** further includes a shaft **250** coupled to the substrate support body **210**. The shaft **250** can be coupled to an actuator (not shown), which can rotate the shaft **250** during processing. The rotation of the shaft **250** can be used to rotate the substrate support body **210** and the substrate **50** positioned on the substrate support body **210** during processing. The rotation of the substrate **50** can improve process uniformity for the process (e.g., deposition) being performed on the substrate **50**.

[0021] The substrate support assembly **200** can further include a heater **280** (e.g., a resistive heater) and a sensor **270**, such as a temperature sensor (e.g., a thermocouple). The sensor **270** can be positioned in an outer region of the substrate support body **210** (i.e., a region not overlying the shaft **250**). As described in further detail below, the sensor **270** can include one or more leads or wired connections routed through an interior **251** of the shaft **250** of the substrate support assembly **200**.

[0022] FIGS. 2A-2C show different stages of a method for forming the substrate support assembly **200** from FIG. 1, according to one embodiment. FIG. 3 is a process flow diagram of a method **3000** for forming the substrate support assembly of FIG. 1, according to one embodiment. The method **3000** is described in reference to FIGS. 2A-2C and FIG. 3.

[0023] The method **3000** begins at block **3002**. At block **3002**, with reference to FIG. 2A, sintering material **201** is placed in a form **60**. The sintering material **201** is generally a ceramic material. In one embodiment, the sintering material **201** is aluminum nitride. The heater **280** is embedded in the sintering material **201**. The heater **280** can include one or more conductors (not shown) that extend out of the bottom of the sintering material, so that electrical power can be provided to the heater **280** after the substrate support assembly **200** is installed in a process chamber. A tube **220** is also embedded in the sintering material **201**. In some embodiments, the tube **220** is embedded in the sintering material **201** at a position below the heater **280**. In some embodiments, the tube **220** can be formed of a material having a high melting point, such as a metal (e.g., molybdenum). For example, the material that the tube **220** is formed of can have a high melting point, such as a melting point that is at least 50° C. higher than the temperature used to perform the sintering process on the sintering material **201** during block **3004** described below. Used herein, a tube generally refers to a structure that fully or partially encloses an interior hollow region. In some embodiments, the tube **220** can be sealed (e.g., capped) at both ends, so that no sintering material enters the tube **220** during the sintering process.

[0024] Furthermore, in some embodiments, an interior **226** of the tube **220** can be filed with a powder **225**. The powder **225** can provide structural support for the tube **220** during the sintering process. The powder **225** can also be formed of a material having a high melting point, such as a melting point that is at least 50° C. higher than the temperature used to perform the sintering process on the sintering material **201** during block **3004**. In some embodiments, the tube **220** and the powder **225** can each be formed of a material having a melting point that is at least 50° C. higher than the melting point of the material used to form the substrate support body (i.e., the sintering material **201**). In some embodiments, the powder **225** can be formed of one or more of carbon, molybdenum, and tungsten. Alternatively, in some embodiments, the interior of the tube

220 is empty during the sintering process during block **3004**. The interior **226** of the tube **220** can be substantially devoid of sintering material **201** during the sintering process, such as less than 5% of the volume of the interior **226** being filled with sintering material **201** during the sintering process.

[0025] At block **3004**, with reference to FIG. 2A, a sintering process is performed on the sintering material **201** in the form **60**. The sintering process is performed with the tube **220** embedded in the sintering material **201**. The sintering process forms the substrate support body **210** as shown in FIG. 2B. The substrate support body **210** includes a top **211**, a bottom **212**, and one or more side surfaces **213** connecting the bottom **212** with the top **211**.

[0026] The sintering process can be performed at a high temperature and a high pressure. In one embodiment, heat is applied to increase the temperature of the sintering material to around 2000° C. Furthermore, in one embodiment, a shaft **75** can lower a plate **70** to apply high pressure to the top of the sintering material **201**, such as a pressure greater than 100 bar, to press the sintering material **201** against the form **60** during the sintering process.

[0027] At block **3006**, with reference to FIG. 2B, a channel **215** can be machined (e.g., drilled) into the substrate support body **210**. The channel **215** can extend from an opening **214** in the bottom **212** of the substrate support body **210** to the interior **226** of the tube **220**. After the channel **215** is formed, the powder **225** can be removed (e.g., vacuumed out) from the interior **226** of the tube **220** through the channel **215**. Alternatively, in some embodiments, the powder **225** can be burned off at a relatively low temperature, for example at atmospheric pressure, to remove the powder **225** from the interior **226** of the tube **220**. Alternatively, in some embodiments, the tube **220** can include an additional portion (not shown) that can extend to the bottom **212** of substrate support body **210**, which can reduce the amount of effort, such as machining (e.g., drilling) used to access the interior **226** of the tube **220**.

[0028] At block **3008**, with reference to FIG. 2C, the substrate support body **210** is attached to the shaft **250**. In some embodiments, the substrate support body **210** is diffusion bonded to the shaft **250**.

[0029] At block **3010**, with reference to FIG. 2C, the sensor **270** is positioned in the interior **226** of the tube **220**. The sensor **270** can be flexible and can be inserted into the interior **226** of the tube **220** through the interior **251** of the shaft **250** and then through the channel **215** of the substrate support body **210** that connects the interior **251** of the shaft **250** to the interior **226** of the tube **220**. The sensor **270** can include a sensing portion **271** that is positioned at location in the interior **226** of the tube **220** that does not overlie the interior **251** of the shaft **250**. For example, the sensing portion **271** can be positioned to underlie an outer zone (e.g., edge region) of the substrate **50** (see FIG. 1) during processing. The sensor **270** can further include one or more connectors **272** (e.g., wires) that extend through the interior **251** of the shaft **250** and electrically connect the sensing portion **271** to another device, such as a controller (not shown) that can receive measurements from the sensor **270** during processing.

[0030] FIGS. 4A-4D show different stages of forming an alternative substrate support assembly **400**, according to one embodiment. FIG. 5 is a process flow diagram of a method **5000** for forming the substrate support assembly **400**, according to one embodiment. The method **5000** is described in reference to FIGS. 4A-4D and FIG. 5.

[0031] The method **5000** begins at block **5002**. At block **5002** with reference to FIG. 4A, a first plate **410** is formed by performing a first sintering process on a first sintering material **415**, and a second plate **420** is formed by performing a second sintering process on a second sintering material **425**. The first sintering material **415** and the second sintering material **425** are generally formed of a ceramic material, such as aluminum nitride. The first sintering process and the second sintering process can generally be performed as described above in reference to FIG. 2A, for example by applying heat and pressure to the sintering material after placing the sintering material in a form, such as the form **60** shown in FIG. 2A. Although the first and second sintering processes can each

be similar to the sintering process described above in reference to FIG. 2A, the first and second sintering processes only partially sinter the first sintering material **415** and the second sintering material **425**. In some embodiments, the first and second sintering processes can be (1) performed for less time and/or (2) performed at lower temperatures and/or pressures than would otherwise be required to more fully sinter the corresponding first sintering material **415** and the second sintering material **425**. Used herein, partial sintering can refer to a sintering process in which the end product of the partial sintering process can be further sintered to reduce to the volume of the end product by at least another 10%, such as by at least another 25% without losing any significant mass. For example, the first plate **410** is formed by a partial sintering process at block **5002**, and the volume of the first plate **410** can be reduced by at least another 10% or at least by another 25% in a subsequent sintering process, such as the process described below in reference to block **5006** when the first plate **410** is joined with the second plate **420**.

[0032] The first plate **410** can include a first side **411** and an opposing second side **412** with one or more sidewalls **413** connecting the first side **411** with the second side **412**. Similarly, the second plate **420** can include a first side **421** and an opposing second side **422** with one or more sidewalls **423** connecting the first side **421** with the second side **422**. The first side of **421** of the second plate **420** can include a recess **430** that can be used to create the space that is subsequently used to insert the sensor **270** into the substrate support body **405** as described in more detail below. Although not shown, the first plate **410** or the second plate **420** can include the heater **280** shown in FIG. 1.

[0033] At block **5004**, with reference to FIG. 4B, the powder **225** is placed in the recess **430** of the second plate **420**. As described above, the powder **225** can be formed of a material having a high melting point, such as the powder **225** being formed of one or more of carbon, molybdenum, and tungsten.

[0034] At block **5006**, with reference to FIG. 4B, the first plate **410** is bonded with the second plate **420**. In some embodiments, the first plate **410** is joined to the second plate **420** by using a sintering process. Because the first plate **410** and the second plate **420** were only partially sintered at block **5002**, the plates **410**, **420** can be joined together at block **5006** by completing the sintering process, for example when the plates **410**, **420** are pressed together at high temperature (e.g., 2000° C.) and high pressure (e.g., >100 bar). A cavity **431** is formed between the first plate **410** and the recess **430** of the second plate **420**. The powder **225** can provide structural support for the cavity **431** during the sintering process.

[0035] The sintering process performed at block **5006** forms the substrate support body **405** shown in FIG. 4C. The substrate support body **405** includes a top **401**, a bottom **402**, and one or more side surfaces **403** connecting the bottom **402** with the top **401**.

[0036] At block **5008**, with reference to FIG. 4C, a channel **406** can be machined (e.g., drilled) into the substrate support body **405**. The channel **406** can extend from an opening **407** in the bottom **402** of the substrate support body **405** to the cavity **431**. After the channel **406** is formed, the powder **225** can be removed (e.g., vacuumed out) from the cavity **431** through the channel **406**.

Alternatively, in some embodiments, the powder **225** can be burned off at a relatively low temperature, for example at atmospheric pressure, to remove the powder **225** from the cavity **431**.

[0037] At block **5010**, with reference to FIG. 4D, the substrate support body **405** is attached to the shaft **250**. In some embodiments, the substrate support body **405** is diffusion bonded to the shaft **250**.

[0038] At block **5012**, with reference to FIG. 4D, the sensor **270** is positioned in the cavity **431** of the substrate support body **405**. The sensor **270** can be inserted into the cavity **431** through the interior **251** of the shaft **250** and then through the channel **406** of the substrate support body **405** that connects the interior **251** of the shaft **250** to the cavity **431**. The sensor **270** can include the sensing portion **271** that is positioned at location in the cavity **431** that does not overlie the interior **251** of the shaft **250**. For example, the sensing portion **271** can be positioned to underlie an outer zone (e.g., edge region) of the substrate **50** (see FIG. 1) during processing. The sensor **270** can

further include the one or more connectors 272 (e.g., wires) that extend through the interior 251 of the shaft 250 and electrically connect the sensing portion 271 to another device, such as a controller (not shown) that can receive measurements from the sensor 270 during processing.

[0039] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof.

Claims

1. A substrate support assembly comprising: a shaft; and a substrate support comprising: a substrate support body attached to the shaft, the substrate support body formed of a first material; and a tube positioned inside the substrate support body, the tube formed of a second material having a melting point that is higher than a sintering temperature of the first material.
2. The substrate support assembly of claim 1, further comprising a sensor positioned inside the tube.
3. The substrate support assembly of claim 2, wherein a sensing portion of the sensor is positioned at location in an interior of the tube that does not overlie the shaft.
4. The substrate support assembly of claim 3, wherein the sensing portion of the sensor is configured to underlie an edge region of a substrate positioned on the substrate support during processing.
5. The substrate support assembly of claim 2, wherein the sensor further includes one or more connectors that extend through an interior of the shaft.
6. The substrate support assembly of claim 1, wherein the substrate support body is formed of a ceramic material and the tube is formed of molybdenum.
7. The substrate support assembly of claim 6, wherein the ceramic material is aluminum nitride.
8. A method for forming a substrate support comprising: embedding a tube in a sintering material, an interior of the tube substantially devoid of the sintering material; performing a sintering process on the sintering material with the tube embedded in the sintering material to form a substrate support body; and inserting a sensor into the interior of the tube after performing the sintering process.
9. The method of claim 8, wherein the sintering material is formed of a ceramic material.
10. The method of claim 9, wherein the tube is formed a material having a higher melting point than a temperature of the sintering process performed on the sintering material.
11. The method of claim 8, wherein the sintering material is formed of aluminum nitride and the tube is formed of molybdenum.
12. The method of claim 8, further comprising attaching the substrate support body to a shaft configured to rotate the substrate support body, wherein the sensor is inserted into the tube in the substrate support body through an interior of the shaft to an interior of the tube by extending through a channel extending from an opening in a bottom side of the substrate support body to the interior of the tube.
13. The method of claim 8, wherein a heater is embedded in the sintering material when the sintering process is performed on the sintering material, and the sensor is a thermocouple.
14. The method of claim 8, further comprising placing a powder in the interior of the tube before performing the sintering process, wherein the powder is in the interior of the tube during the sintering process.
15. The method of claim 14, further comprising removing the powder from the tube after performing the sintering process and before inserting the sensor inside the tube, wherein the powder is formed of one or more of copper, molybdenum, or tungsten.
16. A method for forming a substrate support comprising: forming a first plate by performing a first sintering process on a first sintering material, the first plate having a first side and an opposing second side, wherein the first sintering process partially sinters the first sintering material; forming

a second plate by performing a second sintering process on a second sintering material, the second plate having a first side and an opposing second side, wherein the second plate includes a recess on the first side and the second sintering process partially sinters the second sintering material; placing a powder inside the recess; positioning the first plate on top of the first side of the second plate, wherein a region between the first plate and recess of the second plate forms a cavity; and performing a third sintering process on the first plate and the second plate to join the first plate to the second plate and form a substrate support body having a top, a bottom, and one or more sides connecting the top to the bottom, wherein the powder is formed of a material having a higher melting point than a temperature of the first plate and the second plate during the third sintering process to join the first plate to the second plate.

17. The method of claim 16, further comprising: forming a channel extending through from an opening in the bottom of the substrate support body to the cavity; and removing the powder from the cavity.

18. The method of claim 17, further comprising: inserting a sensor inside the cavity through the channel; and attaching the substrate support body to a shaft configured to rotate the substrate support body, wherein the sensor is inserted into the cavity through an interior of the shaft to the cavity by extending through a channel extending from an opening in a bottom side of the substrate support body to the cavity.

19. The method of claim 16, wherein the first sintering material and the second sintering material are each formed of a ceramic material, and the powder is formed of one or more of copper, molybdenum, or tungsten.

20. The method of claim 16, wherein a heater is embedded in the first sintering material when the first sintering process is performed on the first sintering material
